

Chapter 1: What Is Statistics?

Quantitative Business Methods

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Descriptive and Inferential Statistics

You are a policy analyst of the Bahraini Ministry of Education and you have been asked to do a study on the quality of primary schools in Bahrain.

You collected data on a sample of schools and inputted it into a dataset.

Descriptive and Inferential Statistics

<i>i</i>	School	Type	Score
1	Ain Jaloot Primary Intermediate Girls	Female	Good
2	Abusaiba Primary Boys	Male	Bad
3	Aisha Um Al Moamneen Primary Girls	Female	Good
4	A'ali Primary Boys	Male	Good
5	Sanad Primary Girls	Female	Bad
6	Ahmed Al Fateh Primary Intermediate Boys	Male	Bad
7	Al-Budaiyya Primary Intermediate Girls	Female	Good
8	Hajer Primary Girls	Female	Bad
9	Al Imam Malek Bin Anas Primary Boys	Male	Bad
10	Al Busaiteen Primary Girls	Female	Good

Descriptive and Inferential Statistics

In order to conduct your study, you summarise the information in a small table:

	Good	Bad
Boys	5	5
Girls	8	2

Descriptive and Inferential Statistics

There are a number facts that you can use to describe your data:

- There are a total of 20 schools in your dataset.
- There are 10 boy schools in your sample.
- There are 10 girl schools in your sample.
- 50% of boys schools achieved a good score.
- 80% of girls schools achieved a good score.

Descriptive and Inferential Statistics

This is known as **descriptive statistics**. You are simply looking at the information that was collected and describing the contents.

Descriptive and Inferential Statistics

Now suppose you are interested to find out whether girl schools are *better* than boys schools. One primitive way to doing this is to simply compare the percentage difference of good schools based on gender. We can clearly see that

$$80\% > 50\%,$$

Based on the sample, an analyst can conclude that **all** primary girl schools in Bahrain are higher quality than **all** boy schools.

Descriptive and Inferential Statistics

This analysis is known as **inferential statistics**. You have analysed your sampled data and drawn a conclusion on all schools in Bahrain.

Descriptive and Inferential Statistics

Activity 1:

You currently work as a strategist for Batelco. Explain whether the following tasks are considered **descriptive** or **inferential** statistics:

- (a) The average network engineer works for 1.5 years before leaving the company.
- (b) Based on a sample of customers, 68% said they would not switch their service, which means that most people are happy with the service.

Descriptive and Inferential Statistics

Solution

- (a) The qualifies as **descriptive** statistics, because we are simply talking about the data collected from network engineers.
- (b) The qualifies as **inferential** statistics, since you are analysing the response from a sample of customers.

Sample and Population

An important fact to consider is that your analysis is only based on 20 schools.

Suppose that three other Ministry of Education employees also did their own investigations and collected the following data from 20 schools each:

Sample X

	Good	Bad
Boys	4	6
Girls	9	1

Sample Y

	Good	Bad
Boys	5	5
Girls	7	3

Sample Z

	Good	Bad
Boys	6	4
Girls	8	2

Sample and Population

Even though they all have different numbers, the overall story remains the same.

It seems that based on the sample of schools, each investigator can make an inference that primary girls schools are better than primary boys schools.

These are known as **samples**. They represent parts of the overall total of primary schools across Bahrain.

Sample and Population

Suppose you now go and collect data on **all** 130 primary schools in Bahrain. You find the following information:

	Good	Bad
Boys	53%	47%
Girls	96%	4%

Since you have collected this information from all 130 schools, then we say that this dataset is the **population** of primary schools in Bahrain.

Types of Variables

In order to assess whether a public school is good or bad, you need to base your conclusion a set of variables. For example:

- Funding received from the government.
- Number of teachers and students.
- Number of classrooms
- Classroom size.
- Exam scores.
- Number of air conditioners.

These are examples of **quantitative variables**. That is, the data that you record will be inputted as numbers.

Types of Variables

School	Funding (BD)	Students	Teachers	Class Size	Exam (%)	A/Cs
1	450,000	600	45	24	78	32
2	520,000	750	50	25	84	38
3	380,000	480	35	24	71	25
4	600,000	800	55	25	89	40

Types of Variables

You should be aware that there are two types of quantitative variables: **continuous** and **discrete**.

Types of Variables

Continuous variables can be described as numbers that can value within a range.

- For example, the amount of money received by a school in 2026 can 2.6 Million BD or the average score of the mathematics exam was 67.3%.

Discrete variables can be described as those that can counted.

- For example, a school can be 50 teachers, 100 classrooms and each classroom can have 2 ACs. A school **cannot** have 100.5 classrooms or 50.3 teachers or 0.5 ACs.

Types of Variables

Data can also be **qualitative**. This means that the data collected is some descriptive measure and not something measurable. For example:

- City where the school is located.
- The education degree of the principal.
- If there is a requirement to wear uniforms.

Types of Variables

School	City	Principal's Degree	Uniform Required
1	Manama	Bachelor's	Yes
2	Riffa	Master's	Yes
3	Muharraq	PhD	No
4	Hamad Town	Master's	Yes

Levels of Measurement

Both quantitative and qualitative data can be categorised by four levels of measurement:

Level	Description	Examples
Nominal	Categories with no natural order	Department, industry, school type
Ordinal	Categories with a meaningful order	Satisfaction, credit rating, class rank
Interval	Ordered numeric values with meaningful differences, but no true zero	Celsius temperature, calendar year
Ratio	Ordered numeric values with meaningful differences and a true zero	Salary, sales, enrolment

Levels of Measurement

One way to identify the difference between interval and ratio is to ask if *is zero a genuine absence of the measured quantity?*

Only ratio requires zero to be a proper "zero". For example, 0 BD, 0 employees, 0 students in a classroom, where the zero is properly understood.

However, 0°C does not literally mean "no temperature" and "Year 0" does not literally mean "no years".

Ethics in Statistics

We conclude this chapter by informing students that statistics can easily be manipulated or misrepresented. As a statistician, you must try to collect data in principled way.

- Make sure you collect a proper representative sample.
- Do not omit relevant statistics.
- Do not let bias impact your collection.

Activity 3

Explain how each scenario impacts the school study.

- (a) You only collect data on schools from high-income neighbourhoods.
- (b) You notice that teachers in boy's schools get paid much less compared to girl's schools, but cannot collect the data.
- (c) The principal only lets you interview teachers with high evaluations.

Solution:

- (a) By omitting schools from low-income areas, your sample is not representative of the population. The average scores, performance and ranks of the schools will be higher than what is true.
- (b) Its possible that because teachers in boy's schools get paid less, they are not as good as teachers in girl's schools. This is probably useful information that can explain the difference in performance that you would report to the ministry of education.
- (c) Your data will be biased because you are only getting information from teachers that perform well in the school. Low-performing teachers will not be considered and so you will improperly rank these schools.